

Protocol for a Systematic Literature Review

Interpretability Methods for Large Language Models: Mapping the Field and Assessing Applications to Inference-Time Steering

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1. Background and Rationale

Large language models (LLMs) increasingly produce fluent but occasionally factually incorrect or hallucinated content. A growing body of research investigates the internal representations of LLMs (e.g., hidden states, residual stream activations, sparse autoencoder features) as a source of interpretable signal that could, in principle, be used to explain and potentially steer model behavior. This systematic review aims to map the landscape of interpretability methods for LLMs and to assess the extent to which such methods have been applied to real-time (inference-time) steering of generation, as opposed to purely post-hoc explanation.

This review will follow the PRISMA 2020 methodology, adapted for the software engineering / computer science domain following established guidance (e.g., Kitchenham & Charters) in place of the biomedical PICO framework.

2. Research Questions

RQ1. What interpretability methods exist for large language models (LLMs), at which stage are they applied (training / post-hoc / inference-time), what level of model access do they require (black-box / grey-box / white-box), and what are their main advantages and limitations?

- RQ1a. What interpretability methods for LLMs exist, and at which stage (training / post-hoc / inference-time) are they applied?
- RQ1b. What level of model access (black-box / grey-box / white-box) does each method require?
- RQ1c. What are the main limitations and computational costs of each method?

RQ2. Which LLM interpretability methods that leverage the model's internal signal have already been applied to steer generation directly at inference time, as opposed to post-hoc explanation of already-generated output, and how effective have they been empirically at improving generation quality (reducing factual errors/hallucinations)?

- RQ2a. Which specific methods have been applied to inference-time steering, and what metrics were used to measure their effectiveness?
- RQ2b. What limitations do the authors of these studies report (inference-time computational cost, robustness of the effect, side effects on generation quality)?

3. PICO-Adapted Framework

Component	Definition
Population	Decoder-only transformer large language models ($\geq 1\text{B}$ parameters, or explicitly described as an LLM by the authors)
Intervention	Methods extracting or leveraging an interpretable signal from internal representations

	(hidden states, attention weights, residual stream, neuron activations, sparse autoencoder / transcoder / crosscoder features)
Comparison	Post-hoc, input/output-level explainability methods that do not access internal representations (optional axis, used for contrast in synthesis, not as an exclusion criterion)
Outcome	Classification of methods along the axes defined in RQ1; empirically measured effectiveness of inference-time steering in reducing factual errors/hallucinations (RQ2)

4. Eligibility Criteria

4.1 Inclusion Criteria

- Population: study concerns a decoder-only transformer LLM (≥ 1 B parameters, or explicitly described as an LLM by the authors)
- Intervention: study proposes or analyzes a method extracting signal from internal representations (hidden states, attention weights, residual stream, neuron activations, SAE/transcoder/crosscoder features)
- Publication period: 2022–2026
- Publication type: peer-reviewed publication OR arXiv preprint (publication status recorded as a separate field during data extraction — see Section 6)
- Language: English
- Full text: openly accessible
- Empirical content: includes at least minimal empirical validation of the proposed/analyzed method

4.2 Exclusion Criteria

- Model is not a transformer-based LLM (classical RNN/LSTM, or models < 1 B parameters without explicit justification of relevance)
- Method analyzes only input/output (pure prompt engineering, input-attribution without access to internal representations)
- Published before 2022
- Not in English
- Full text not accessible despite institutional subscription attempts (e.g., IEEE Xplore items without open-access copy)
- Duplicate of a more complete version of the same work (only the most complete/final version is included)
- Survey/review papers on interpretability in general (tracked separately as related work for the Introduction, not included in data extraction)
- Blog posts / technical notes without minimal technical detail and not formatted as an arXiv preprint

5. Information Sources and Search Strategy

Databases to be searched: ACM Digital Library, IEEE Xplore, ScienceDirect, arXiv. Scopus was excluded due to lack of institutional access at the time of the review (ITMO University does not hold a full Scopus subscription); ScienceDirect was used as an alternative source with comparable disciplinary coverage. Semantic Scholar was excluded from the final search due to substantial content overlap with the other four databases (particularly arXiv, which Semantic Scholar largely indexes) combined with resource constraints; this is noted as a limitation of the review.

5.1 Database Selection Changelog

- Removed: Scopus — no institutional subscription available at ITMO University (identified after attempting the search; only Scopus Preview, without full document search or Advanced Search, was accessible).
- Added: ScienceDirect — selected as a replacement for Scopus, providing comparable multidisciplinary coverage and accessible through the ITMO institutional subscription.
- Removed: Semantic Scholar — excluded from the final search strategy due to substantial content overlap with the four retained databases (it largely aggregates records already indexed by arXiv and the other sources) and to keep the manual search-execution workload feasible within the project timeline.

Final Boolean search string (Population block deliberately excludes "transformer*" to reduce noise from unrelated transformer applications, e.g. computer vision, audio, RL; borderline cases will be assessed manually during title/abstract screening):

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("large language model*" OR "LLM*") AND ("interpretab*" OR "explainab*" OR "mechanistic interpretability" OR "probing" OR "steering vector*" OR "activation steering" OR "sparse autoencoder*" OR "transcoder*" OR "crosscoder*" OR "representation engineering" OR "concept activation vector*" OR "attention attribution" OR "saliency" OR "feature attribution" OR "chain-of-thought faithfulness" OR "CoT faithfulness")
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A pilot search will be conducted to verify that the search string retrieves a set of known seed papers relevant to the topic before finalizing and applying the search across all sources.

5.2 Database-Specific Query Adaptation

- ScienceDirect: Advanced Search, field set to "Title, abstract, keywords", using the same boolean structure without the TITLE-ABS-KEY() wrapper (Scopus-specific syntax not applicable)
- IEEE Xplore: ("Abstract": "large language model*" OR "Abstract": "LLM*") AND ("Abstract": "interpretab*" OR ...)
- ACM Digital Library: constructed via the Advanced Search field builder (per-field terms), following the same logical structure
- arXiv: executed via the arXiv API (export.arxiv.org/api/query), using field-prefixed boolean queries (abs:...) with a submittedDate range filter; results retrieved in batches and converted to RIS for import into the reference manager

6. Data Extraction

Data extraction will be performed only for studies that pass full-text screening (Section 4). For each included study, the following fields will be extracted into a structured spreadsheet:

Field	Description
Citation key	First author, year
DOI / URL	Direct link to the publication
Venue	Journal, conference, or "arXiv preprint"
Publication status	Peer-reviewed / preprint
Method name	Name of the technique as given by the authors
Object of analysis	Attention / hidden states / residual stream / neurons / SAE features
Type of internal signal	Scalar / vector / distribution / sparse code
Level of model access	Black-box / grey-box / white-box
Stage of application	Training-time / post-hoc / inference-time
Granularity	Token-level / layer-level / model-level

Applied to steering	Yes/No; if yes, how
Evaluation metrics	Names of metrics used
Measured effect	Concrete quantitative/qualitative result (e.g., magnitude of hallucination reduction)
Datasets / benchmarks	Datasets or benchmarks used for evaluation
Limitations (as reported by authors)	Free text
Extracted by	Human / LLM-assisted (model name, version, date) + human verification
Notes / interpretation	Free-text notes for synthesis and gap analysis

Given resource constraints, the first pass of data extraction will be LLM-assisted: a large language model will populate the fields above from the full text of each included study, after which every entry will be manually reviewed and corrected by the reviewer. Agreement between the LLM's first-pass extraction and the reviewer's final corrections will be assessed on a pilot subsample (10–15 studies) to identify fields prone to systematic disagreement, which will then receive closer manual attention across the full sample. This process, including the specific model and version used, will be reported transparently in the Methods section of the resulting article.

7. Synthesis Method

Results will be synthesized in three steps, performed after the extraction table (Section 6) is complete:

- Tabular/matrix classification: cross-tabulation of studies across the axes defined above (object of analysis × access level × application stage × granularity), using pivot tables.
- Timeline visualization: publication counts per method family by year, to identify growing and declining sub-areas.
- Gap analysis: comparison of RQ1 coverage (the full map of interpretability methods) against RQ2 coverage (methods applied to inference-time steering) to identify empty or sparsely populated cells in the classification matrix, which will motivate the proposed thesis method.

8. Amendments

Any changes to the research questions, eligibility criteria, or search strategy made after this protocol is registered will be documented as a dated amendment, referencing this original protocol version, including the date and rationale for the change.

Amendment 1 (v1.1 → v1.2)

Date: 2026-09-19

Changes: (1) Scopus removed from the database list due to lack of institutional access; ScienceDirect added as a replacement. (2) Semantic Scholar removed from the database list due to substantial content overlap with the remaining four databases and resource constraints. Final database set: ACM Digital Library, IEEE Xplore, ScienceDirect, arXiv.

Amendment 2 (v1.2 → v1.3)

Date: 2026-09-19

Changes: Title/abstract screening was conducted with LLM assistance. A large language model (Claude Sonnet 5, Medium reasoning effort) was used alongside the reviewer to apply the inclusion/exclusion criteria in Section 4 to each record; the reviewer verified all LLM-flagged decisions and independently checked a subsample for reliability, consistent with the LLM-assisted extraction methodology already described in Section 6. Following title/abstract screening, 1,502 records were advanced to full-text eligibility assessment.

As a calibration step prior to completing full-text screening of the remaining records, the reviewer purposively read a subset of 51 full texts, selected after reviewing their abstracts, to refine practical judgment on what constitutes a substantive contribution to RQ1 and/or RQ2 before applying this judgment to the remaining pool. This subset was not selected through a formal random sampling procedure; its role is calibration of the full-text screening approach, not final study selection.

Amendment 3 (v1.3 → v1.4)

Date: 2026-09-26

Changes: Following title/abstract screening, 1,502 records were advanced to full-text eligibility assessment. This pool exceeded what could be read and analyzed within the project timeline, and the eligibility criteria in Section 4 had to be tightened for the full-text stage. A full-text eligibility criterion was defined, in addition to the criteria in Section 4, requiring records to meet all three of the following conditions to be included in the review: (1) Non-instrumentality — the interpretability method is the study's own methodological contribution, not a borrowed tool applied to an unrelated application domain (e.g., autonomous driving, molecular editing, model compression, code vulnerability detection); (2) Inference-time intervention — the internal signal actively changes, corrects, or steers model output during generation, as opposed to pure diagnosis/analysis, training-time weight updates, or one-off architectural decisions made prior to deployment; (3) Strict RQ2 outcome — the measured effect is specifically factual accuracy or hallucination/error reduction, as opposed to other controllable behaviors (e.g., sentiment, personality, emotion, ideology, semantic consistency) that do not directly address RQ2 as formulated.

This criterion was applied to all 1,502 records via LLM-assisted classification (chat-based, processed in batches). The reviewer first manually classified 10 records spanning all three outcome categories and used them as worked few-shot examples to guide the LLM's classification of the remaining records; of these 10, 3 met all three conditions. The LLM then classified the remaining 1,492 records, of which 71 were found to meet all three conditions. In total, 74 records (71 LLM-classified plus 3 from the reviewer's few-shot set) met all three conditions and were included in the review. The remaining 1,428 were excluded at the full-text stage, with the specific unmet condition(s) recorded as the exclusion reason for each record.

— *End of protocol* —